

9/22/2026

Assignment-4

Prepaid By
~PINTU MER

1. **Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.**

➤ **RAM Reseating – Steps**

1. Shut down the computer and **unplug the power cable**.
2. Open the computer/laptop back cover.
3. Locate the **RAM slot** on the motherboard.
4. Push the **clips on both sides** of the RAM module outward.
5. Carefully remove the RAM module.
6. Check the RAM and slot for **dust or damage**.
7. Align the notch on the RAM with the slot.
8. Insert the RAM firmly until the clips **click into place**.
9. Close the cover and reconnect the power.
10. Turn on the computer and check whether it **boots normally**.

Issues observed: No visible damage or major issue was found. The RAM was reseated successfully and the system booted normally.

2. **Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Hint: Think about display issues, driver checks, and physical connections.**

➤ **GPU Troubleshooting Checklist**

1. **Check the problem** – See if there is no display, screen flickering, or game crashing.
2. **Check GPU connection** – Make sure the GPU is properly inserted in the PCIe slot.
3. **Check power cable** – Make sure the GPU power cable is connected properly.
4. **Check display cable** – Make sure the HDMI/DisplayPort cable is connected correctly.
5. **Check GPU driver** – Update or reinstall the GPU driver.

6. **Clean the GPU** – Remove dust from the GPU fan and check if it is overheating.
 7. **Test the PC** – Run a game and check if the problem is gone.
 8. **Confirm the fix** – If the display and games work normally, the GPU problem is fixed.
3. **Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**

Storage Issue

➤ steps

1. **Turn off the PC** and unplug the power cable.
2. Open the PC case and **disconnect the SATA cable** from the HDD/SSD.
3. Turn on the PC and enter **BIOS/UEFI**.
4. Check the storage list. The **HDD/SSD will not be detected**.
5. You may see an error like **“No Boot Device Found”** or **“Operating System Not Found.”**
6. Turn off the PC and **reconnect the SATA cable** properly.
7. Turn on the PC and check BIOS again.
8. The **HDD/SSD should appear** in the storage list.
9. Boot the computer and confirm that the operating system starts normally.

Observation: Before reconnecting, the drive was not detected. After reconnecting, the drive was detected and the PC booted normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

➤ **Random PC Restart – Troubleshooting**

- **Component I would check first: RAM**

Reason: Faulty or loose RAM can cause random restarts, crashes, or freezing.

❖ **Diagnostic Steps**

1. **Turn off the PC** and unplug the power.
2. Open the PC case and **check the RAM**.
3. Remove the RAM and **clean the contacts/slot** carefully.
4. **Reinsert the RAM firmly** until the clips lock.
5. If there are two RAM sticks, **test one stick at a time**.
6. Run **Windows Memory Diagnostic** to check for RAM errors.
7. Use the PC for some time and check if the **random restart happens again**.

Conclusion: If the PC works normally after testing/reseating the RAM, the RAM was likely the problem.